Abeer Sethia

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PROFILE

AI Research Engineer with experience in **real-time medical imaging, trajectory reconstruction, anatomical detection, and volumetric segmentation**. Developing YOLO-based fetal anatomical detection for live ultrasound scans at Origin Medical Research Lab. Strong foundation in computer vision and medical image analysis, with hands-on experience in PyTorch, PyTorch3D, ITK, MONAI, and PyDICOM.

EDUCATION

Manipal Institute of Technology, Manipal 2022–2026
Bachelor of Technology - Data Science and Engineering, Minor in Multimodal Intelligent Systems *CGPA: 7.52/10*
Relevant Coursework: Computer Vision, Deep Learning, Natural Language Processing, Statistics, Mathematical Foundations, Image Processing

TECHNICAL SKILLS

- **Medical Imaging:** CT, MRI, Ultrasound, DICOM Processing, 3D Volumetric Segmentation, Anatomical Detection
- **3D Geometry & Visualization:** SE(3) Transformations, Coordinate Geometry
- **Computer Vision & Deep Learning:** CNNs, U-Net, Object Detection, Attention Mechanisms & Transformers, Self-Supervised Learning
- **Medical Imaging Frameworks:** PyTorch3D, ITK, MONAI, PyDICOM, OpenCV, 3D Slicer
- **Frameworks & Infrastructure:** PyTorch, TensorFlow, CUDA, Docker, Git, AWS EC2, AWS S3
- **Languages:** Python, C++, Bash (Linux), SQL, LaTeX

PROFESSIONAL EXPERIENCE

AI Research Intern, **Origin Medical Research Lab** September 2026–Present
- Developing a real-time fetal ultrasound examination assistant using **YOLO-based computer vision models** for 2nd and 3rd trimester anatomical detection during live ultrasound scans.
- Working with **AWS EC2 and S3** to support cloud-based model development, computational workloads, and medical imaging data storage.

Technical Intern, **Philips Healthcare** March–July 2025
- Developed and optimized a medical image segmentation framework for liver lesions, evaluating and fine-tuning **nnU-Net, MedSAM, and MedSAM2** on a clinically validated dataset of 500+ CT/MRI scans.
- Achieved **0.92 Dice and 0.89 IoU**, outperforming standard U-Net baselines by 12%.

RESEARCH EXPERIENCE

Research Intern – AI, UTSAAH Lab, **Indian Institute of Science Bangalore** December 2025–July 2026
- Developed a self supervised trackerless freehand 3D ultrasound trajectory reconstruction pipeline predicting frame-to-frame relative **SE(3) rigid-body transformations** from short B-mode image sequences.
- Integrated geometric consistency constraints to reduce trajectory drift and improve spatially consistent reconstruction of ultrasound acquisition trajectories.

Research Intern – AI, KLIV Lab, **Indian Institute of Technology Kharagpur** April–October 2025
- Engineered a temporal CNN autoencoder to reconstruct 3D chaotic attractor dynamics from a single-channel observable, achieving **98.3% phase-space correlation in X/Y and 93.4% in Z** at an 8:1 compression ratio.

SELECTED PROJECTS

HVQ-AP: Hippocampal Volume Quantification for Alzheimer's Progression GitHub
- Developed an end-to-end 3D MRI processing pipeline for hippocampal volumetric analysis, integrating **DICOM ingestion using PyDICOM, 3D CNN segmentation, and longitudinal volume tracking**.
- Achieved **91.6% Dice and 84.6% Jaccard** on the Medical Segmentation Decathlon benchmark, enabling quantitative analysis of anatomical structures from volumetric medical images.

DropEdge++ with Johnson–Lindenstrauss Projection
- Developed a graph neural network framework combining topology-aware edge dropout with randomized dimensionality reduction to mitigate oversmoothing and improve computational efficiency.
- Achieved up to **8% accuracy gains and 3× faster training** on the Cora citation network benchmark.

ACHIEVEMENTS

- **Medical Imaging Hackathon Winner:** Awarded "Best Research Methodology" at the Origin Medical Research Labs Medical Imaging AI Hackathon, along with a cash prize.

CO/EXTRA CURRICULAR

- **Expertise Head – Research Society Manipal:** Mentored 20+ students in research and scientific writing.